

Predicting the Outcome of a Soccer Match Using Machine Learning Models

Abstract

Prediction of the outcome of a sports game is a favourite pastime for sports fans and researchers. The interest has intensified in recent years due to data availability, the development and successful implementation of machine learning algorithms, and the proliferation of internet gaming. This research will focus on developing a machine learning (ML) or artificial intelligence (AI) model for accurately predicting match outcomes using publicly available game results data for the English Premier League. The development efforts will include experimentation using weather data and constructs such as fatigue and momentum. Ensemble techniques such as stacking or voting are also explored to produce a final model that produces useful and accurate results.

Keywords:

Sports Betting, Machine Learning Models, XGboost, Random Forest, Support Vector Machine, Convolutional Neural Network, Ensemble Technique.

1. Introduction

Predicting the outcome of a sports game is a favourite pastime for sports fans and researchers. With the advent of the internet and the proliferation and availability of game data, research on how to use quantitative techniques and, more recently, machine learning or artificial intelligence algorithms to improve prediction has intensified. Interest is also fueled by increased sports betting through various gaming websites.

It is well recognized that a profitable betting strategy relies on the optimal use of odds data from the bookmakers and accurate predictions of match outcomes. Following the footsteps of recent research, as identified in the following section, this research will focus on developing a machine learning (ML) or artificial intelligence (AI) model for the accurate prediction of match outcomes. The research will use publicly available game results data for the English Premier League to train various models for predicting the outcome of a soccer match in the League.

The research's main contributions are twofold. We will experiment with using weather data, in addition to game data and additional features such as fatigue and momentum, to create useful predictive models. On the other hand, the research deploys several machine learning algorithms to develop initial prediction models. Some of these models will then be combined using a stacking or voting approach to arrive at a final model that produces more accurate results. We believe that using this approach is promising in predicting game outcomes.

This paper is organized as follows: Section II gives an overview of research on predicting soccer game outcomes and other sports outcomes. Section III presents the methodology used in this research. Section IV presents results using several machine learning models and ensemble techniques. Section V includes discussions of the results and some suggestions on work for the future. We close with a brief Conclusion section.

2. Existing Work

The use of ML algorithms for the development of sports betting strategies has proliferated in the last three years. For example, these papers [1, 2, 3, 4, 5, 6, 7, 8, 9] were published in 2023. Several overviews in this regard can also be found in [10] or [8]. The survey paper by Tokic et al. [11] is also useful, though it was published in 2022.

2.1. Using Machine Learning Models for Predicting Soccer Game Outcomes

Using ML algorithms to predict the outcomes of a soccer match dates back to the mid-2010's. For example, using a data set extracted from 110 games played in the 2014-2015 English Premier League (EPL) season with features that include Home and Away goals, Home and Away shots, Home and Away corners, Home and Away Odds, Home and Away attack strength, Home and Away Players' performance index, Home and Away Managers' performance index, Home and Away managers' win, as well as Home and Away streak, Igiri et al [12] in 2014 implemented a logistic Regression (LR) and an Artificial Neural Network (ANN) model to predict the outcome of 20 matches in 2014-2015. Results of the research showed that an accuracy rate (not explicitly defined in the paper) of 85% with ANN and 93% with LR is possible. One interesting aspect of the approach by the authors was using a genetic algorithm to "calculate the weights of the features" of the data set before input to ML algorithms.

Constantinou in 2019 [13] described an ambitious project to build Dolores, "a model designed to predict football match outcomes in one country by observing football matches in multiple other countries." This model used a mixture of the dynamic ratings and the Hybrid Bayesian Networks approach to train a single dataset that incorporates match outcomes from 52 football leagues worldwide and was used to predict 206 future match outcomes from 26 different leagues from March 31 to April 9 in 2017. Dolores placed 2nd in the international special issue competition, Machine Learning for Soccer.

Carlioni et al in 2021 [14] presented the design and implementation of ML models for predicting the outcome of soccer matches for sports betting. Data were extracted from the Web through a scraping process and used as input to a comprehensive suite of ML algorithms such as Logistic Regression (LR), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes (NB), Random Forest (RF), and a Four-layer Artificial Neural Network (ANN). Based on the criteria of accuracy (not clearly defined in the paper), the ANN model developed is the best for the over-under 2.5 market. Further experiments also showed "encouraging performance in terms of the Return on Investment (ROI) criteria."

Wheatcroft and Sienkiewicz [15] also 2021 examined the need to calibrate probability forecasts of soccer matches produced by various ML models. The authors found that it is essential to evaluate the performance of the models with and without calibration to ensure that the extra step is indeed required. Using the forecasts and two betting strategies, the authors demonstrated how the hyperparameters of ML models could be selected to optimize the decision of betting or not betting forecasts.

Nivetha et al. [16] in 2022 explored using a deep learning framework to forecast soccer outcomes. The project team used a Long Short Term Memory (LSTM) algorithm, a modified version of a Recurrent Neural Network (RNN) for this purpose. Using data from the "International Football results from 1872 to 2018", "FIFA World Cup 2018", and the "FIFA Soccer Rankings" data sets, Rahman [17] used a similar approach, that is an LSTM deep learning network, but a different data set to predict the outcome of the FIFA World cup 2018 group stage matches and

achieve an accuracy rate of 63.3%. The author believes that “this accuracy can be increased with proper datasets and more accurate information of the teams. For more accurate performance of the prediction, prior and more information about each team, player and match is desirable.”

Peters and Pacheco [18] used line-up statistics of teams in the EPL from 2020 to 2022 to build heuristic and ML models to predict soccer scores. The research found that the Support Vector Regression (SVR) algorithm outperforms others in predicting final scores. Moreover, the researchers managed to “predict correctly all relegated teams after forecast 100 consecutive matches” and using the forecasts was profitable (42% return). Interestingly, the researchers concluded that “lineups do not improve predictions”.

A Convolutional Neural Network (CNN) is a special deep learning algorithm designed for classification and other pattern detection tasks such as image classification. Several research projects in the last three years have used this algorithm to predict sports event outcomes and soccer games. Wagenaar et al. [19] is the first research team to use the CNN algorithm in the sports prediction arena. Instead of predicting the outcome of a soccer match, the team uses a “deep CNN” to predict goal-scoring opportunities in soccer using position data from “29 matches played by a German Bundesliga team.” While their ideas are very interesting, they cannot be used directly to predict the outcome of a soccer match.

On the other hand, Chen [20] used three typical machine learning algorithms, CNN, RF, and SVM, to develop models to predict the result of a football match based on a “player ability index” derived from data found in the website of the International Federation of Association Football (FIFA). The accuracy of these three methods is between 54% and 58%. The researcher argued that these results “are acceptable since they are all higher than the prediction accuracy of the famous football analyst of BBC, Mark Lawrenson, which is only 52%.” In addition, the accuracy of the convolution neural network is higher than the prediction accuracy of the authoritative football gambling organization Pinnacle Sports, which is only 55%.” In any case, the research concluded that the four-layer CNN algorithm produced the best results.

Randrianasolo [21] used a CNN to develop models for predicting the 2020 men’s EURO and 2022 women’s EURO, resulting in a 69.8% accuracy and an 80% accuracy, respectively. The research claimed that these results represent a 2% improvement and a 10% improvement over those using an ensemble technique approach in predicting the outcomes of these two soccer tournaments.

Based on these results, the CNN algorithm seems to be a viable alternative for developing models to predict the outcome of a soccer match. The other side of the coin would be the structure of the network and the selection and engineering of features that should be fed to the algorithm.

For women’s soccer, Grollet al. [22] used several ML algorithms to predict the FIFA Women’s World Cup 2019 outcome.

2.2. Feature Selection

The choice of features used as input to the ML models for predicting game outcomes is highly diverse. Depending on the game itself, the league within which the game is played, and the ML algorithms employed, the set of features could be very different.

As mentioned, Chen used a player ability index as input to the predictive models [20]. Peters et al. [18] analyzed “the role of lineups in the final scores using machine learning prediction models” they developed. Historical data and machine learning models were employed to develop these models for predicting the outcome of English Premier League (EPL) games in 2020 to

2022. Through the analyses of features used, they found that statistics on goalkeepers are more critical than those for the attackers in predicted goals scored. Two other findings of interest are that data on lineups do not improve predictions and that Support Vector Regression outperforms other algorithms in predicting final scores. and that lineups do not improve predictions.

Rose et al. [23] explored different Machine learning techniques to predict the score and outcome of soccer matches using custom-generated features. The features are used to train the selected algorithms.

According to Zimmermann et al., [24], features are more important than the algorithms used in predicting the outcome of an NCAA basketball game. This sentiment is echoed in the paper by Iskandaryan et al. [25], which stated that accurate prediction of soccer game outcomes depends on the features included as much as the algorithm used. For example, the authors pointed to the work by Kampakis and Adamides [26] that showed that the integration of Twitter could significantly improve the final results. Iskandaryan et al. then analyzed several research studies ([27, 28, 29, 30, 31, 32, 33]) that related weather conditions to different features of a sports game, such as successful passes, injuries, etc. From these studies, the authors used data from five seasons of Spanish La Liga and Segunda division games between 2013–2014 to 2017–2018 and the weather data to answer the research question: “How do weather conditions affect the results of soccer matches?” [25]

Another research question the authors attempted to answer is: “which ML methods can predict the results of soccer matches with greater accuracy, taking the weather conditions into account as a principal feature?” [25]. The study concluded that including weather-related variables significantly increased the accuracy of the predictions in the implemented models.

2.3. Machine Learning Models for Predicting Game Outcomes in Other Sports

ML has also been used to build models for predicting outcomes of other sports. Beal et al. [34] built and evaluated nine algorithms using data sets of 1280 games over five seasons to predict outcomes of National Football League (NFL) games. Their experiments showed that the Naive Bayes algorithm was the best classifier with an accuracy of 67.5% and an F1 Score of 0.67. The team believed this algorithm performs the best when the features used are independent and dependencies between features are similar. Juuri [35], Roumani et al. [36], and Patel [37] did the same for the NFL, etc.

Hubacek et al. in 2019 [38] incorporated the bookmaker’s predictions into their prediction models by reducing correlation as an optimization tool. The models were developed by implementing a CNN with a convolution layer that “enabled to leverage a vast number of player-related statistics as its input.” Combining with elements of modern portfolio theory, the authors yielded positive cumulative profits in experiments with NBA data. More recently, Liu [39] in 2020, Chen et al. [40] in 2021, Alonso et al. [41] in 2022, and Liu et al. [42] used ML technique to predict NBA game outcomes.

Interestingly, professional baseball teams have used a quantitative approach to manage the games for many years [43]. However, few studies have used ML to predict game outcomes in Major League Baseball [44, 45].

2.4. Model Evaluation

Chicco et al. [46] show the advantage of using the Matthews correlation coefficient (MCC), first introduced by Matthews [47], over the F1 score and accuracy in binary classification evaluation. There are many references in this regard; see, for example, [48].

3. Methodology

Given the problem’s symmetry, we will focus on predicting a home team’s win or loss. Therefore, our problem could be viewed as classifying the outcome of a match into one of these two states: Win (W) or Not Win (NW) of the home team by an ML model developed using the data from previous seasons.

3.1. Data Collection and Processing

Data on soccer, especially for those on the EPL, is widely available. For this research, we have chosen the data set that is made available for free from the [49]. Data on matches in the 2019 to 2022 seasons was downloaded and used in the modelling process.

Unlike Iskandaryan et al. [25] reviewed above, the research team speculated that the use of the weather-related variables associated with the home field of the game and without considerations of those associated with the home field of the opposition team might have an impact on the outcome of the matches. As weather forecasts are readily available, they could be used to predict match outcomes if their inclusion improves the accuracy of the prediction. Accordingly, weather data associated with the soccer matches was also downloaded from [50] (variables 24-29 listed below):

3.2. Exploratory Data Analysis

Before splitting the data set into the training and testing data sets, we performed an exploratory data analysis and used the results for feature engineering.

Figure 1 shows the distribution of the home team wins over the 2019-2021 seasons. It confirms that the data set is not severely imbalanced and we do not need to pay particular attention to this problem.

3.3. Feature Engineering

The extracted data sets described in the last section were transformed to form the training and testing data set for model development:

1. Match Date (dd/mm/yy) - identifier; not used in modelling
2. Home Team - identifier; not used in modelling
3. Opposition Team - identifier; not used in modelling
4. Time of match kick-off (afternoon, evening)
5. Game result (W, NW)
6. Home Team Goals (not used in the model)
7. Opposite Team Goals (not used in the model)
8. Home Team Previous Game (W, NW)
9. Opposition Team Previous Game (W, NW)
10. Home Team Previous Game Goals
11. Opposition Team Previous Game Goals
12. Home Team Previous Game Shots
13. Opposition Team Previous Game Shots
14. Home Team Previous Game Shots on Target
15. Opposition Team Previous Game Shots on Target

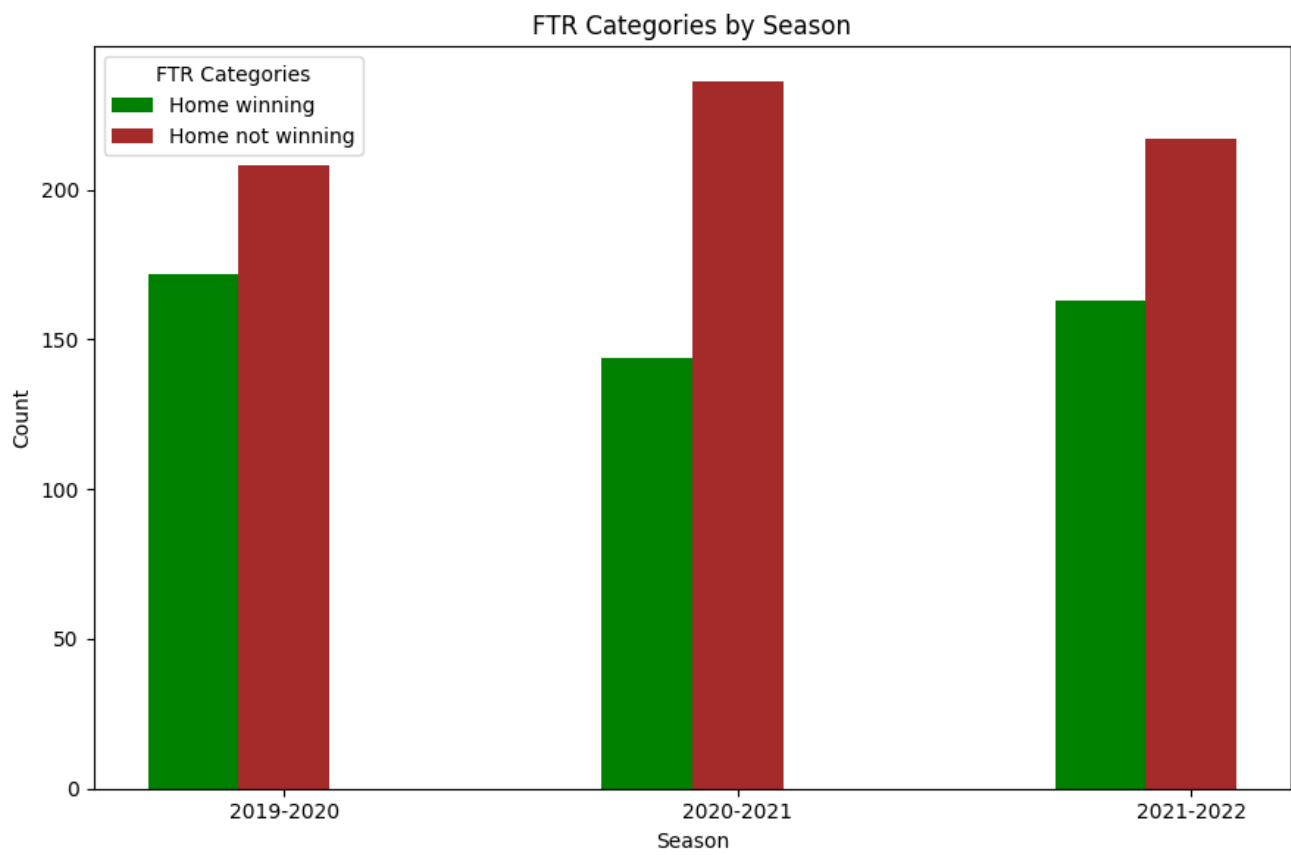


Figure 1: Home Wins - 2019 to 2021

16. Home Team Previous Game Corners
17. Opposition Team Previous Game Corners
18. Home Team Previous Game Fouls Committed
19. Opposition Team Previous Game Fouls Committed
20. Home Team Previous Game Yellow Cards
21. Opposition Team Previous Game Yellow Cards
22. Home Team Previous Game Red Cards
23. Opposition Team Previous Game Red Cards
24. Temperature (degrees Celsius)
25. Relative Humidity (%)
26. Precipitation (mm)
27. Wind Direction (degrees)
28. Wind Speed (m/s)
29. Weather Conditions (e.g., Clouds, Rains, Clear, etc.)

Variable 5) is the target variable of the models. Intuitively, fatigue should be considered in any predictive model of game outcome. A more rested team should have a higher chance of winning. The question remains in which proxy variable(s) should be used to represent the fatigue factor. For this study, we use the following variables as proxy for the fatigue factor:

- Time between the previous and current game
- Average number of days of elapsed time between the last three games

Similarly, many fans and sports analysts believe in the impact of momentum on a team's winning, for example, Steeger et al. in hockey [51]. Again, the challenge is in defining and selecting feature(s) to represent momentum. Our first attempt in this regard is as follows:

- Average number of goals in the last three previous home team games
- Average number of goals in the last three previous opposition team games
- Average number of shots in the last three previous home team games
- Average number of shots in the last three previous opposition team games
- Average number of shots on target in the last three previous home team games
- Average number of shots on target in the last three previous opposition team games
- Average number of corners in the last three previous home team games
- Average number of corners in the last three previous opposition team games
- Average number of fouls in the last three previous home team games
- Average number of fouls in the last three previous opposition team games
- Average number of yellow cards in the last three previous home team games

- Average number of yellow cards in the last three previous opposition team games
- Average number of red cards in the last three previous home team games
- Average number of red cards in the last three previous opposition team games
- Count number of win games in the last three previous home team games
- Count number of win games in the last three previous opposition team games

The numerical features were standardized before building the models. We use two one-hot variables to represent the kick-off time and eight one-hot variables to represent the weather condition. Together, 52 variables are used in the modelling process.

To fully evaluate the potential of using historical data and a machine learning algorithm to predict match outcomes, data on bookmaker odds should not be included as features for the models [38]. Instead, we will contextualize our prediction results with the bookmakers' predictions through the odds.

3.4. Training and Testing Data

The data set's first two seasons (2019-2021) will be used to train the ML models using the algorithms discussed below. Data from the third season (2022) will be used as the testing data set.

3.5. Machine Learning Algorithms

Four basic ML algorithms, Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), and XGboost (XGB), were considered in the research. They are selected for their differences in structure and approach in the modelling process. An excellent overview of using some of these basic algorithms in sports betting can be found in [52]. The LightGBM algorithm [53] has provided good predictive results (see, for example, Sari et al. [54]), and the Bayesian Optimization option of the LightGBM algorithm is also included in the study.

As documented above, a Convolution Neural Network (CNN) for predicting soccer and other sports events has been implemented in several research projects, for example, Randrianasolo [21], Chen et al. [20] or Wagenaar et al. [19]. This algorithm is also used in this study to build a prediction model to gauge whether a more advanced neural network structure would improve predictive accuracy.

Stacking and voting are ensemble techniques used to combine selected classifiers once they are developed to improve performance. For example, the stacking technique was used by Jaiyeoba et al. [55] to increase classification accuracy in classifying skin diseases. See also the use of stacking in detecting diabetes [56] and in education [57]. Similarly, the voting ensemble technique was used recently in Atitallah et al. [58] for detecting IoT malware. Other recent applications of this technique can be found in [57], kuman et al. [59], and Farooqi et al. [60]. Both techniques were implemented in the Scikit Learn package in Python ([61] and [62]) and are therefore readily available. We will explore the use of these two techniques in this research.

3.6. Performance Evaluation

The popular measurement of performance in classification is based on a confusion matrix. The matrix compares the number of predictions for each class that are correct and those that are incorrect. There are four numbers to pay attention to:

- True positives (TP) - The number of positive observations (home wins) the model correctly predicted as positive (home wins).
- False-positive (FP) - The number of negative observations (non-home Win) the model incorrectly predicted as positive (home wins).
- True negative (TN) - The number of negative observations (non-home Win) the model correctly predicted as negative (non-home Win).
- False-negative (FN) - The number of positive observations (home wins) the model incorrectly predicted as negative (non-home Win).

Using these four numbers, we define the following performance measures for the classifier.

3.6.1. Accuracy

The overall accuracy of a model is simply the number of correct predictions divided by the total number of predictions. An accuracy score will give a value between 0 and 1, a value of 1 would indicate a perfect model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

3.6.2. Precision

Precision measures how good the model is at correctly identifying the positive class. In other words, out of all predictions for the positive class, how many were actually correct? Using this metric alone to optimize a model would minimize the false positives.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

3.6.3. Recall

The percentage of Recall measures how well a model predicts all the positive observations in the dataset correctly. However, it does not include information about false positives.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

Typically, the two percentages, precision and recall, are evaluated through a precision-recall curve.

3.6.4. F1 Score

From the formula below, we can see that the F1 score is defined as the harmonic mean of precision and recall:

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall} \quad (4)$$

As constructed, the F1 score takes a value between zero and one and, therefore, could also be expressed as a percentage. If precision and recall are 1, the F1 score is also 1. On the other hand, the F1 score is 0 when either precision or recall is 0.

3.6.5. Normalized Matthews Correlation Coefficient

In addition to the traditional measures defined above, our research also uses the normalized Matthews correlation coefficient (UMCC) as a measurement tool [47] giving its advantages as outlined in [46]. It is defined as follows

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{A}} \quad (5)$$

$$UMCC = \frac{MCC + 1}{2} \quad (6)$$

Here A is calculated as:

$$A = (TP + FP)(TP + FN)(TN + FP)(TN + FN) \quad (7)$$

Putting the classifier into perspective, we should also examine the performance of bookmakers. This examination is presented in the Discussion section.

4. Results

This section presents the results from the models built with the machine learning algorithms discussed in Section III. With the exception of the LR and CNN models, they were optimized using a grid search approach. Applying the models to the testing data set, the results are summarized in Table I.

Table 1: Results of the Machine Learning Models

Cell	LR	RF	SVM	XGB	LightGBM	CNN
TP	58	41	106	39	58	50
TN	157	169	106	179	159	159
FP	44	32	95	22	42	42
FN	91	108	43	110	91	99
UMCC	59.3%	57.1%	61.9%	59.9%	59.9%	57.1%
Accuracy	61.4%	60.0%	60.6%	62.3%	62.0%	59.7%
Precision	56.9%	56.2%	52.7%	63.9%	58.0%	54.3%
Recall	38.9%	27.5%	71.1%	26.2%	38.9%	33.6%
F1	46.2%	36.9%	60.6%	37.1%	46.6%	41.5%

The confusion matrix shows that SVM has the highest accuracy rate, behaves very differently from other classifiers, and deals with the slight imbalance issue more effectively. LR turns out to be a good classifier overall. On the other hand, the CNN model does not stand out as an effective classifier. This leads us to implement the stacking and hard voting (Voting) algorithms using the LR, SVM, and LightGBM (GBM) models. The results below confirm our decision

Table 2: Stacking and Voting as Ensemble Predictors

Cell	LR	SVM	GBM	Stacking	Voting
TP	58	106	58	117	80
TN	157	106	159	84	148
FP	44	95	42	117	53
FN	91	43	91	32	69
UMCC	59.3%	61.9%	59.9%	60.7%	63.9%
Accuracy	61.4%	60.6%	62.0%	57.4%	65.1%
Precision	56.9%	52.7%	58.0%	50.0%	60.2%
Recall	38.9%	71.1%	38.9%	78.5%	53.7%
F1	46.2%	60.6%	46.6%	61.1%	56.7%

As expected, the hard voting algorithm combining LR, SVM, and LightGBM produces the best overall results.

5. Discussions and Future Work

As explained in [38], a highly accurate model is not as helpful if it coincides with those used by the bookmakers. On the other hand, an excellent predictive model should achieve a similar level of accuracy as those from the bookmakers. Therefore, it is important to compare any model developed with those articulated by the odds posted by the bookmakers. We used the matches in the testing data set and chose the odds produced by Bet365, William Hill (WH), Pinnacle (PS), and Market overall as a starting point.

To perform the analysis, we convert the odds produced by each bookmaker into a “prediction” for each match in the testing: the bookmaker gives a prediction of a home win if the odd for the home win is lowest among the three odds (home team win, away team win, and tie). With this definition, the interesting result from the Table is that the bookmakers have a high level of agreement. More interesting is the fact that the accuracy rate of the bookmakers is only slightly higher than 65% achieved by the hard voting algorithm. On the other hand, the bookmakers are far superior in the recall rate and hence the F1.

Note that Bet 365, William Hill (WH), and Pinnacle (PS) are three of the largest bookmakers in England.

Given that the bookmakers are consistent in predicting the outcomes of the games in terms of home wins, we will choose Best 365 as the representative bookmaker and compare it with the best ML classifier in Table IV.

Interestingly, the hard voting classifier is better at predicting home losses but less effective at predicting home wins. This is likely due to the slight imbalance of the training data set. In any case, we can conclude that there is considerable disagreement between the ML classifier and the predictions made by the bookmakers. From a betting perspective, this should give rise to profitable opportunities. Work for the future should, therefore, be as follows:

Table 3: Prediction by Major Bookmakers and Hard Voting

Cell	Voting	Bet 365	WH	PS
TP	80	124	125	124
TN	148	109	109	109
FP	53	92	92	92
FN	69	25	24	25
NMCC	63.9%	69.0%	69.4%	69.0%
Accuracy	65.1%	66.6%	66.9%	66.6%
Precision	60.2%	57.4%	57.6%	57.4%
Recall	53.7%	83.2%	83.9%	83.2%
F1	56.7%	67.9%	68.3%	67.9%

Table 4: Actual VS Bet 365 VS Hard Voting Predictions

Games	Count
Actual Home Loss	201
Bet 365 Predicted Home Loss	109
Hard Voting Predicted Home Loss	93
Hard Voting Predicted Home Win	16
Bet 365 Predicted Home Win	92
Hard Voting Predicted Home Lost	55
Hard Voting Predicted Home Win	37
Actual Home Win	149
Bet 365 Predicted Home Loss	25
Hard Voting Predicted Home Loss	18
Hard Voting Predicted Home Win	7
Bet 365 Predicted Home Win	124
Hard Voting Predicted Home Loss	51
Hard Voting Predicted Home Win	73
Total Number of Games	350

1. Conduct further feature engineering experiments regarding momentum and fatigue. Identify and evaluate other features that could be brought into the modelling process
2. Make improvements to the classifiers focusing on re-structuring the CNN algorithm and further experimental with stalking and voting.
3. Formulate a betting strategy taking into account the concurrence and discrepancies between the bookmaker(s) and the prediction

6. Conclusions

In this research, we develop several ML models to predict the outcome of soccer games in the English Premier League. We introduce novel features such as momentum and fatigue as part of the model development process. Two ensemble approaches to combining classifiers were also used to arrive at the final predictive model. The prediction results are comparable to those of the bookmakers in terms of overall accuracy. It is recommended that further work be done to improve the models and develop a betting strategy based on them.

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